Unit 18 - Database Design
Assignment 2 - Theory and database queries

Introduction

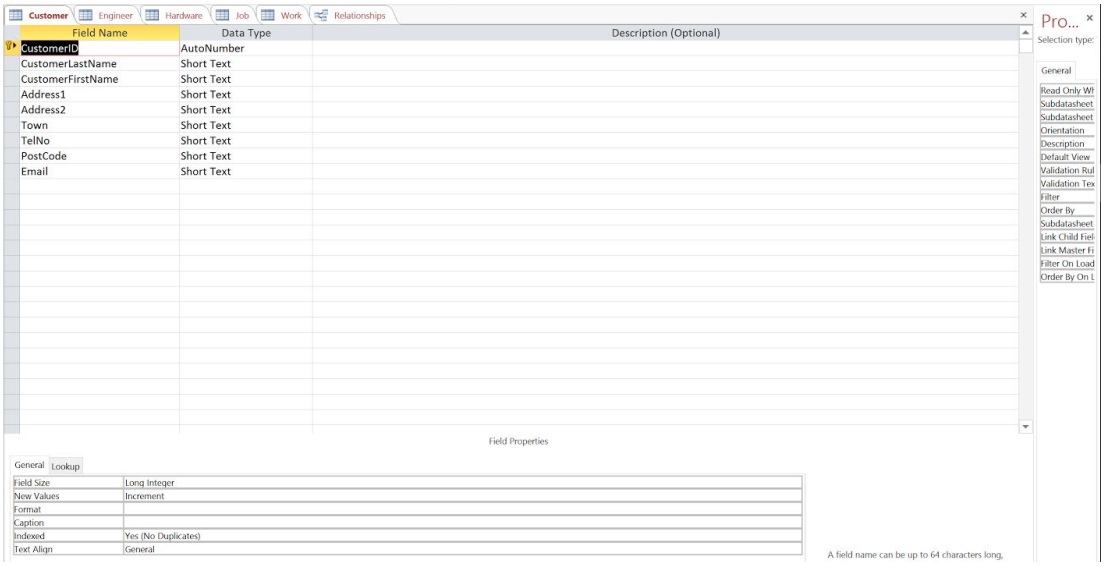
Scenario / Vocational Context: *Highland Hardware Repairs (HHR) is a small company that repairs computers and other IT hardware. Having only been in business for a few months, they have not yet automated the tracking of jobs and are relying on a mix of email, spreadsheets and paper based job sheets to track each repair. However, they have a good idea of what they want, as documented on the next page.*

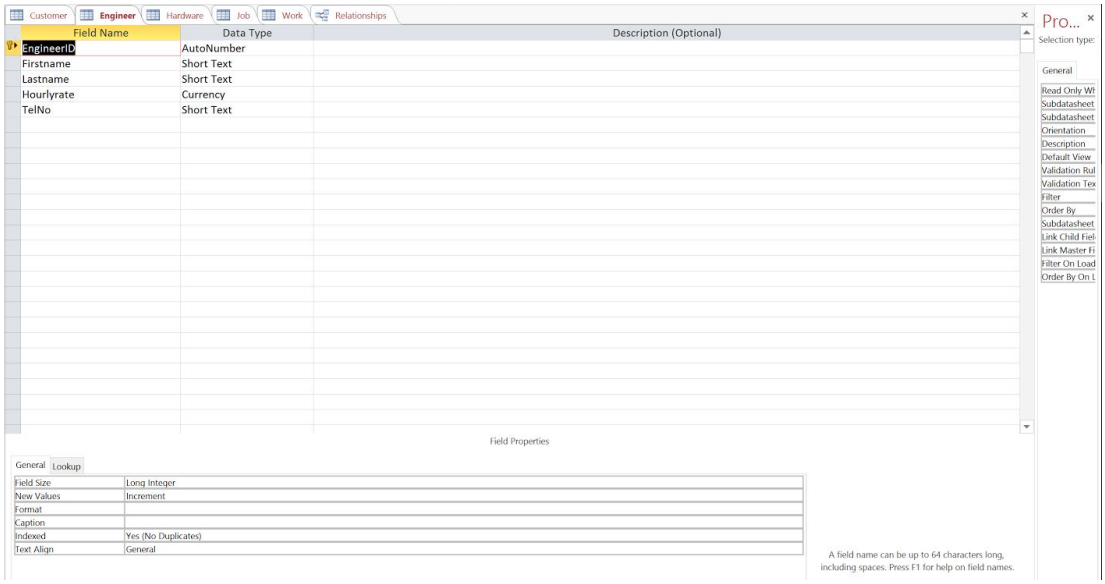
Task 1

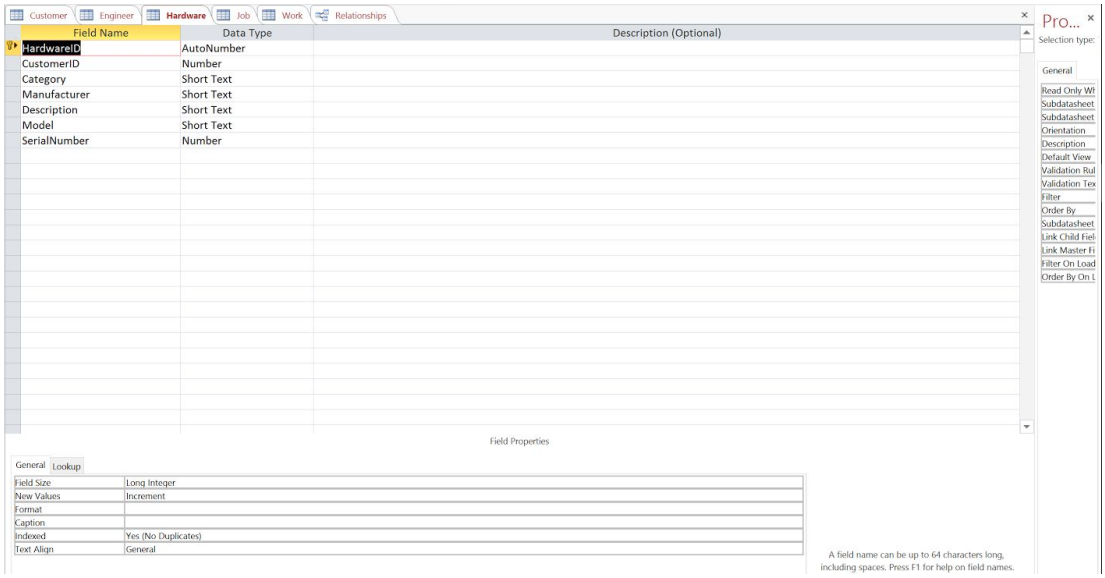
(U18:P3) - Create and populate a database

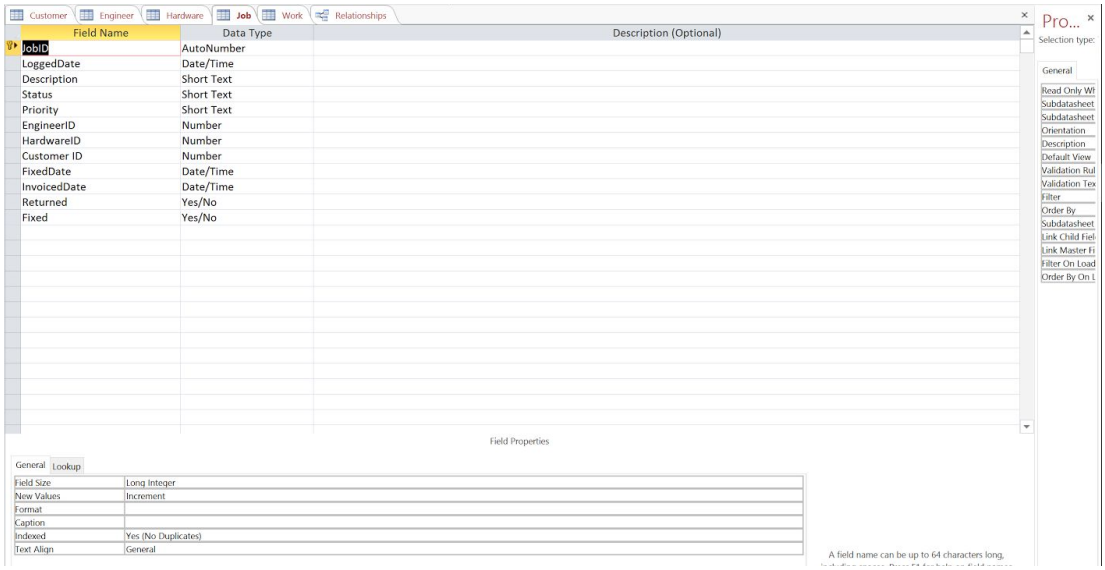
Task description: *In your documentation, show:*

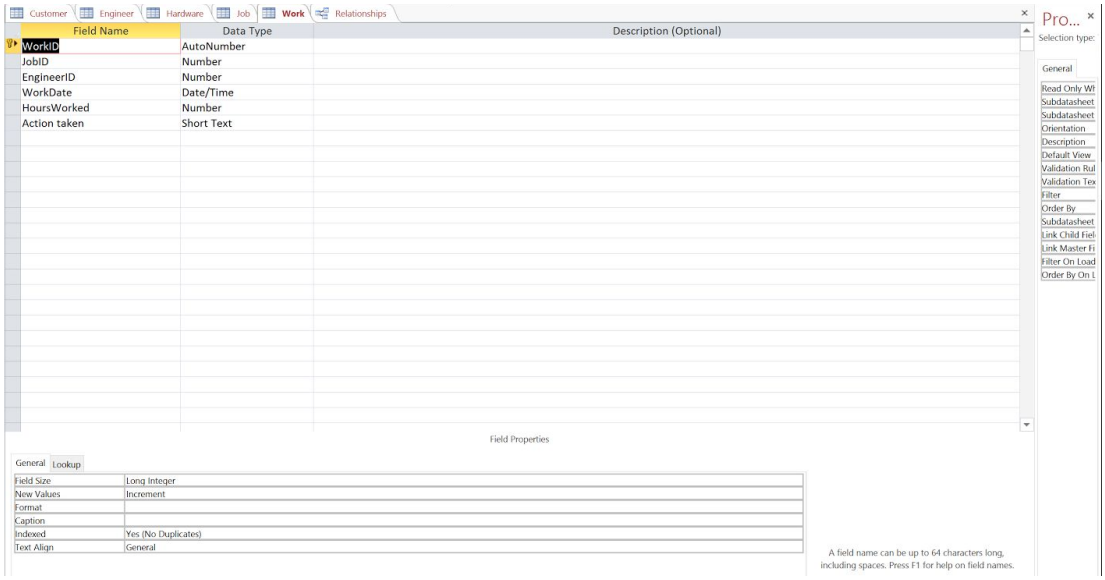
Screen-prints of each table in design view.




Customer
Engineer
Hardware
Job
Work
Relationships
Field Name
Data Type
Description (Optional)
EngineerID
AutoNumber
Firstname
Short Text
Lastname
Short Text
Hourlyrate
Currency
TelNo
Short Text
Field Properties
General
Lookup
Field Size
Long Integer
New Values
Increment
Format
Caption
Indexed
Yes (No Duplicates)
Text Align
General
A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.
Pro...
Selection type:
General


Customer
Engineer
Hardware
Job
Work
Relationships
Field Name
Data Type
Description (Optional)
HardwareID
AutoNumber
CustomerID
Number
Category
Short Text
Manufacturer
Short Text
Description
Short Text
Model
Short Text
SerialNumber
Number
Field Properties
General
Lookup
Field Size
Long Integer
New Values
Increment
Format
Caption
Indexed
Yes (No Duplicates)
Text Align
General
A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.
Pro...
Selection type:
General





Screen-prints of each table showing the records you have entered.

Customer | Engineer | Hardware | Job | **Work** | Relationships

WorkID	JobID	EngineerID	WorkDate	HoursWorke	Action taken	Click to Add
1	1	2	08/03/2017	8	The repairs were simple due to it being a replacable part.	
2	2	7	10/03/2017	6	The parts have arrived with the repairs still in progress.	
3	3	1	16/02/2017	7	The hardware has worn out to a point of unrepairable damage.	
4	4	10	01/02/2017	9	The hardware has been repaired.	
5	5	3	03/03/2017	2	The parts in order for a repair is still in progress.	
6	6	5	10/02/2017	1	The parts have arrived with the repairs still in progress.	
7	7	8	17/02/2017	5	The parts have arrived with the repairs still in progress.	
8	8	4	12/04/2017	6	The hardware is unrepairable due to heavy damage.	
9	9	6	01/03/2017	2	The hardware was binned due to the damage.	
10	10	9	20/01/2017	10	The hardware is beeing repaired.	
11	1	3	17/02/2017	6	The repairs were simple due to it being a replacable part.	
12	2	6	10/03/2017	8	The parts have arrived with the repairs still in progress.	
13	3	8	01/02/2017	5	The hardware has worn out to a point of unrepairable damage.	
14	4	10	03/03/2017	7	The hardware has been repaired.	
15	5	1	03/03/2017	5	The parts in order for a repair is still in progress.	
16	6	5	10/02/2017	7	The parts have arrived with the repairs still in progress.	
17	7	2	09/02/2017	10	The parts have arrived with the repairs still in progress.	
18	8	4	16/02/2017	6	The hardware is unrepairable due to heavy damage.	
19	9	9	06/03/2017	11	The hardware was binned due to the damage.	
20	10	7	22/02/2017	6	The hardware is beeing repaired.	

Customer | Engineer | Hardware | Job | Work | Relationships

CustomerID	CustomerLas	CustomerFirs	Address1	Address2	Town	TelNo	PostCode	Email	Click to Add
1	Peterson	Joseph	16 road	Queen's Road	Saint Helier	07700342423	JE2 5XD	P.Joseph@gmai	
2	Valski	Viktor	74 street	Rue de la Petite Falaise	Trinity	07700532432	JE7 7RX	V.V1990@hotm	
3	Smith	Jonathan	32 road	St.John's Road	Saint Helier	07700745754	JE1 9PA	Smith1980@gm	
4	Manson	David	11 street	Round Street	Saint Peter	07700432544	JE6 8QP	Mason1960@h	
5	Harley	Dylan	67 road	Rte du Marais	Saint Ouen	07700457233	JE4 6TF	DylanH.@gmail	
6	Page	Alfred	54 street	La Rue Jutize	Grouville	07700237823	JE9 2RT	AlfredPage@jer	
7	De Jesus	Andrew	43 street	Saint Marcus Street	Saint Helier	07700763423	JE5 5YZ	DeJesus@gmail	
8	Da Silva	Max	12 road	Tower Road	Saint Helier	07700785761	JE6 9IP	MaxDS@hotma	
9	Warwick	Henry	32 street	Rue du Port	Saint Mary	07700658743	JE4 1YA	Warwick.H@gm	
10	Johnson	Richard	57 street	La Rue de la Capelle	Saint Ouen	07700756438	JE4 7OP	Johnson1980@	

Customer | **Engineer** | Hardware | Job | Work | Relationships

EngineerID	Firstname	Lastname	Hourlyrate	TelNo	Click to Add
1	Hanz	Eugen	£20.00	07700587343	
2	Isabella	Smith	£23.00	07700443228	
3	Olivia	Stingray	£24.00	07700745352	
4	Peter	Gill	£17.00	07700785643	
5	Sophia	Carlson	£26.00	07700184593	
6	Rohan	Kishabe	£28.00	07700885635	
7	Karl	Eisenbahn	£24.00	07700649293	
8	Charlie	Smithson	£26.00	07700436128	
9	Lily	Holley	£19.00	07700647352	
10	Peter	Flanel	£18.00	07700684523	

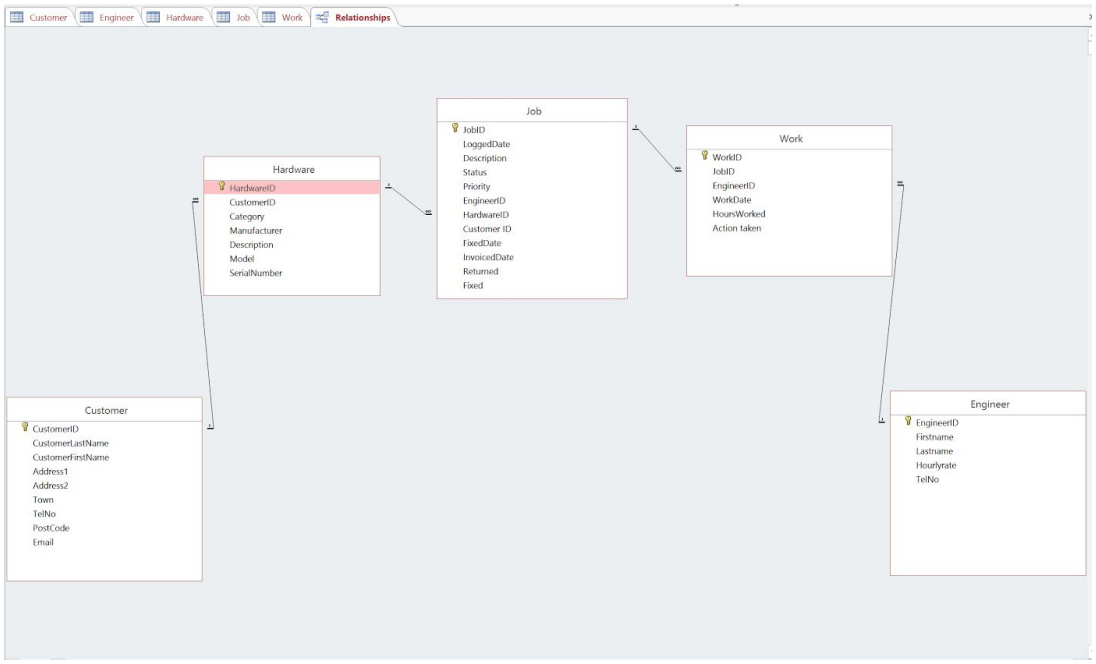
Customer | Engineer | **Hardware** | Job | Work | Relationships

HardwareID	CustomerID	Category	Manufacture	Description	Model	SerialNumbe	Click to Add
1	4	Monitor	ASUS	Monitor used for students	PBQ2412	43633425	
2	3	Laptop	ACER	Laptop reserved for students	laptop 2017	23523423	
3	1	Printer	HP	Printer used in the main corridor	LazerPrnt123	84723934	
4	10	Monitor	DELL	Monitor used for staff	E2016	23746237	
5	7	Tablet	Microsoft	Tablet used for digital graphics teaching	Surface pro 4	54653853	
6	1	Graphics Adapter	AMD	Graphics adapter used on college computers	Radon 4000	43534523	
7	6	Monitor	ACER	Other monitor used for students	PRK342421	34243423	
8	8	Printer	ACER	Other monitor used for staff	PRNT31	55477653	
9	9	Laptop	Packard Bell	Laptop reserved for staff	2016	67546454	
10	2	Tablet	Apple	Tablet used for digital graphics students	Air 2	12846429	
11	5	Monitor	BENQ	Monitor used for graphics design	BQ4322	32432423	
12	5	Graphics Card	Nvidia	Graphics card used for graphics desing	GTX 750	72346667	
13	9	Tablet	Samsung	Tablet used for admins	galaxy tablet 2	65474575	
14	8	Hard drive	Samsung	Hard drive used for college computers	HDD1000	32423678	

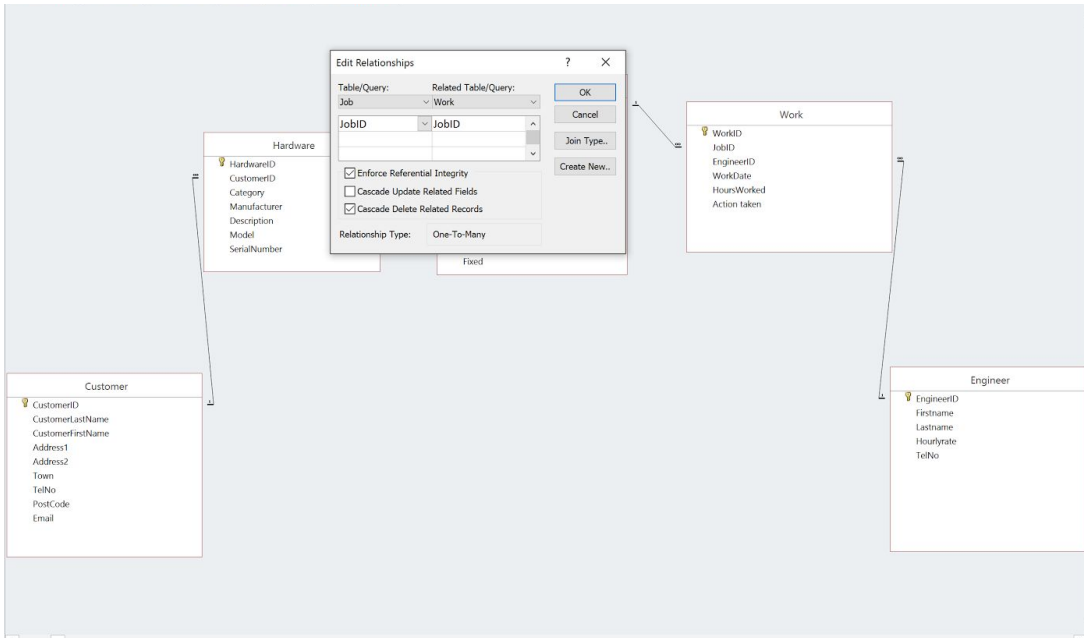
Customer | Engineer | Hardware | **Job** | Work | Relationships

JobID	LoggedDate	Description	Status	Priority	EngineerID	HardwareID	Customer ID	FixedDate	InvoicedDate	Return
1	17/02/2017	This hardware was in need of new parts in order to repair.	Fixed	High	2	10	2	08/03/2017	08/03/2017	☑
2	10/03/2017	This hardware is in need of new parts in order to repair.	In progress	Medium	7	7	6		10/03/2017	
3	01/02/2017	The hardware seems to have heavy damage.	Unrepairable	Low	1	5	7		16/02/2017	☑
4	03/02/2017	Simple repairs were needed in order to get the hardware workin	Fixed	High	10	14	8	01/02/2017	01/02/2017	
5	03/03/2017	This hardware seems like it is in need of new parts.	Awaiting Parts	Medium	3	9	9		03/03/2017	
6	10/02/2017	This hardware seems like it is in need of new parts in order to re	In progress	Low	5	2	3		10/02/2017	
7	09/02/2017	This hardware seems like it is in need of new parts.	Awaiting Parts	Medium	8	1	4		17/02/2017	
8	16/02/2017	The hardware seems to have heavy damage that looks unrepaire	Unrepairable	High	4	12	5		12/04/2017	☑
9	06/03/2017	The hardware has been binned due to it being faulty.	Binned	Low	6	4	10		01/03/2017	
10	22/02/2017	This hardware is being repaired at the moment this was recorde	In progress	High	9	6	1		20/01/2017	

A screen-print of the relationships created (Database Tools ribbon > Relationships button) showing that all relationships have been enforced (all have one to infinity shown).



A screen-print of the relationship between jobs and work records showing that cascade delete has been turned on.



Task 2

(U18:M4) - Implement an automated function

Task description - *Macros or VBA code used in your database to aid navigation and / or*

validation.

Your macros or VBA code is documented and explained in your documentation

In my database, I have included a macros to create 3 forms of data validation and buttons to navigate throughout the forms. I created these functions so the data can't be confused due to a lot of data linking to different tables within the database, and that it would make navigation of the database easier for the user.

For the validation methods, I made 3 different validation methods for my database. Each one has a somewhat different validation so that the data has a less chance in being wrong or invalid.

For the first validation method, I created a few validation rules which has "Like "###########"" written, this is so that only 11 numbers are accepted. The reason that I made this validation rule is due to the fact that telephone numbers consist of 11 numbers. So if there was data for the telephone number which was above or below 11 numbers, the data would be incorrect.

For the second validation method, I created a few validation rules which has "Like "?*@?*.??*"" written, this is so that only a few characters with "@" and "." in between them are accepted. The reason that I made this validation rule is due to the fact that emails consist of a few characters with "@" and "." in between them. So if there was data for the email which didn't consist of few characters with "@" and ".", the data would be incorrect.

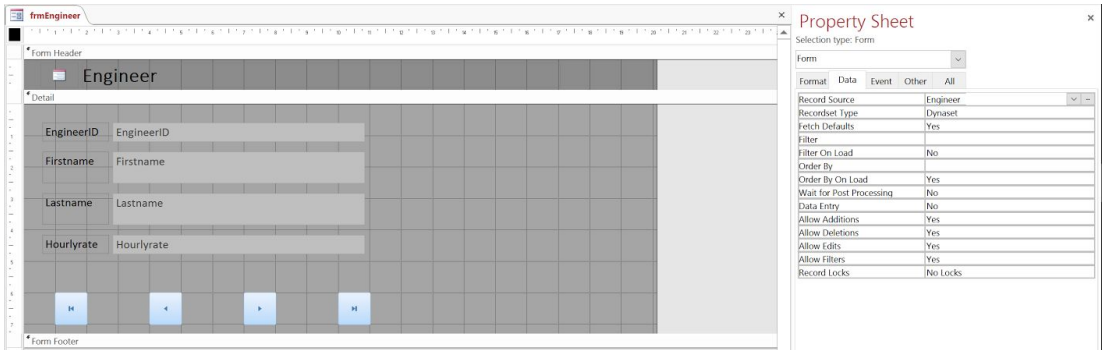
For the third validation method, I created a few lookups which are connected to the queries called "Customer Query" and "Engineer Query", this is so that only the data stored in the query is accepted. The reason that I made this validation method is due to the fact that the customers and engineers data cannot be made up when already created. So if there was data which was meant to store the existing customers or engineers was made up, the data would be incorrect.

For the button navigation, I created a few buttons which are on the forms, this is so that the user can navigate through the forms to see the connected data. The reason that I made this is due to the fact that it makes it easier to look at all the connected data without going through the struggle of reading through all the tables. So if there was no buttons on these forms, the user would have to read through all the tables to see the connected data.

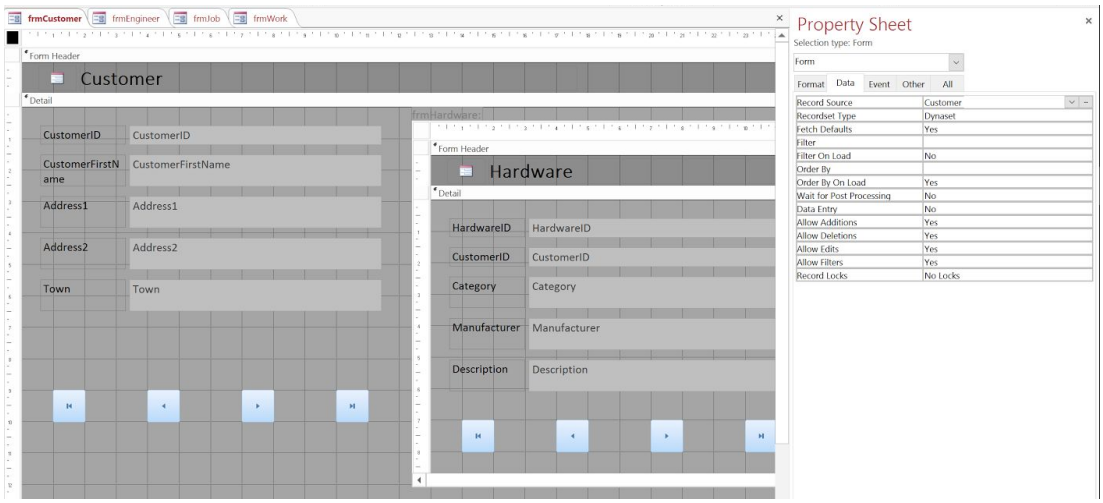
In conclusion, I made these in order to aid navigation and validation of data so that the customer using the database has an easier time accessing and implementing data. I feel that this form of validation and automated function is important to have in my database due to the fact that the customer wouldn't be able to use the database properly without them.

Document your two forms by providing annotated screen-prints that show and explain:

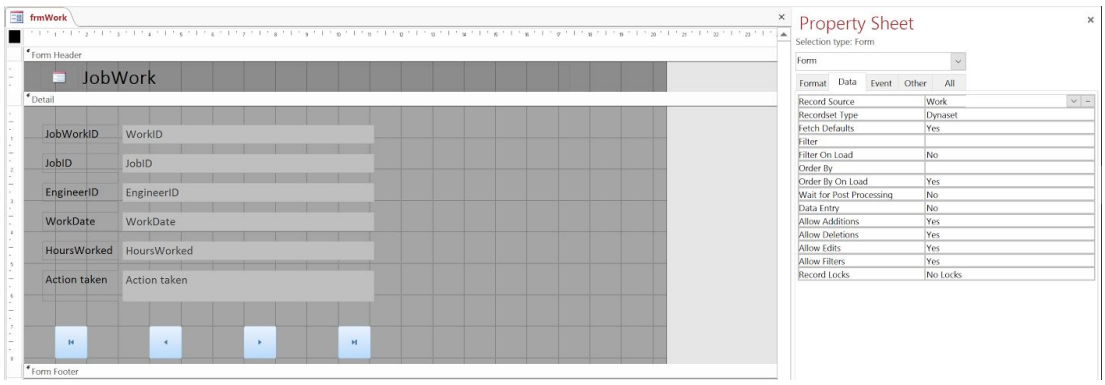
Your completed forms in design view.



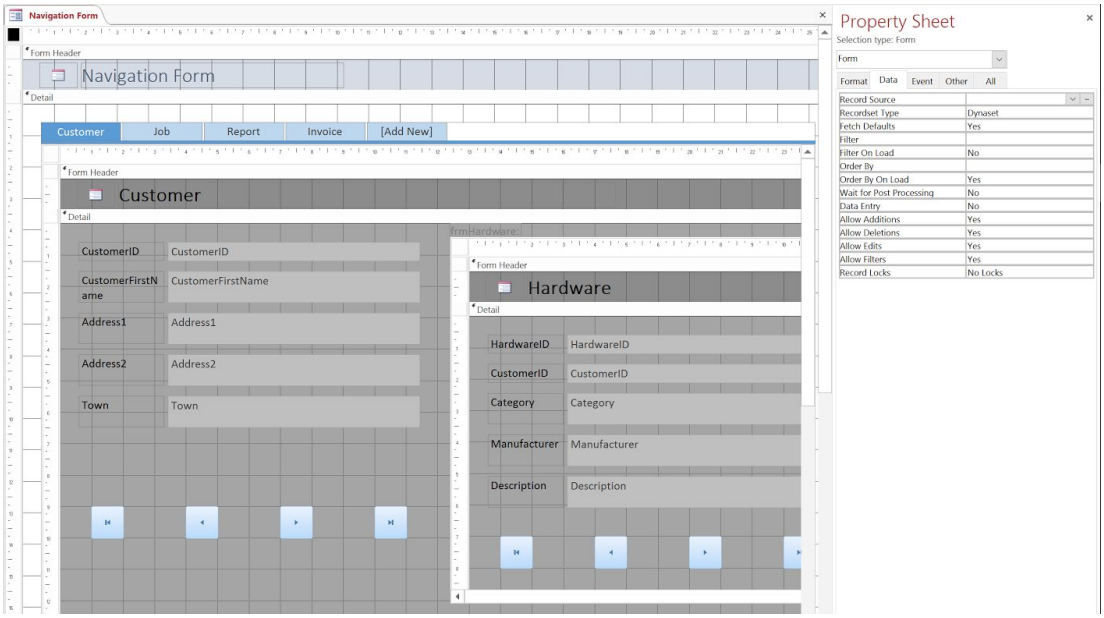
This is the design view of the Engineer form.

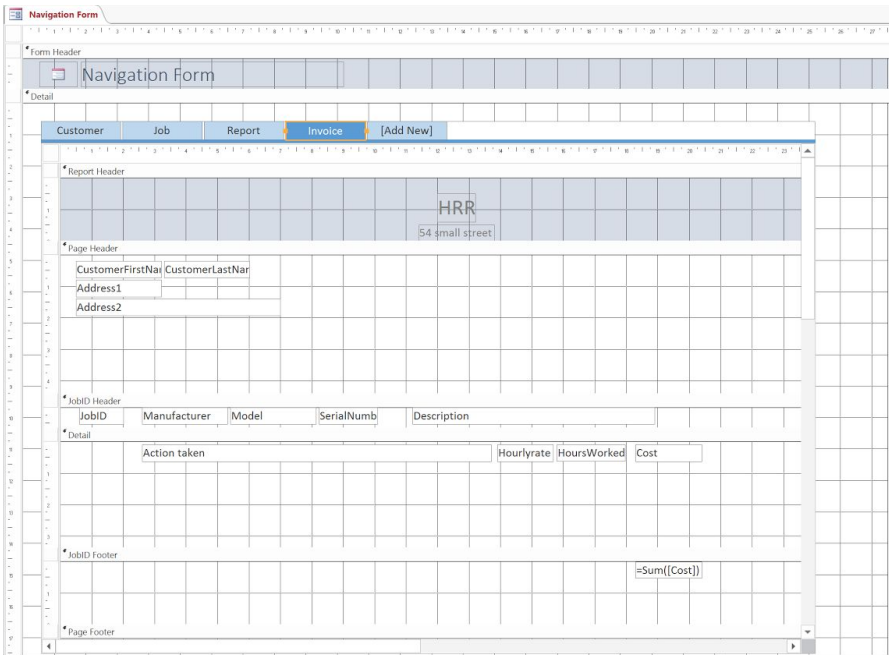


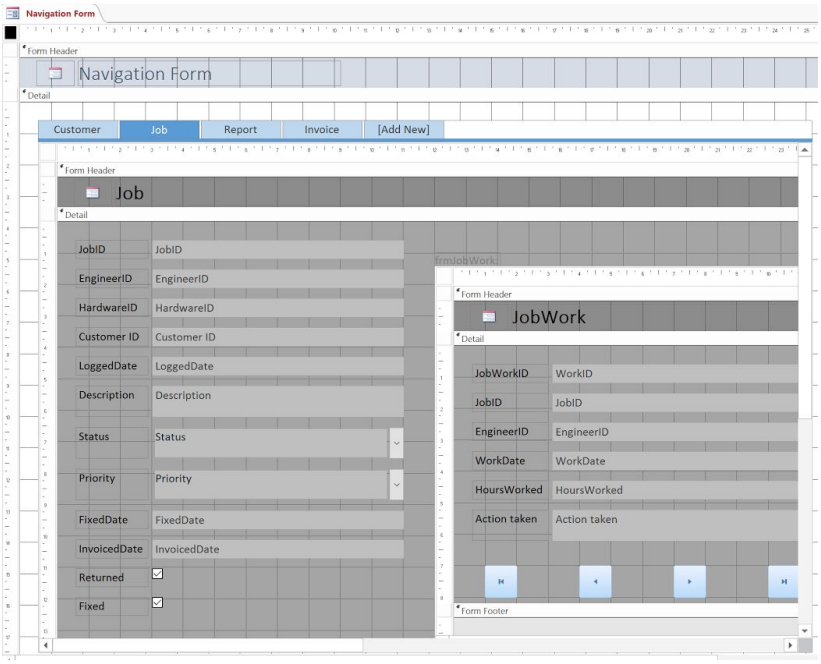
This is the design view of the Customer form, which has the Hardware form embedded within the Customer form, giving easier view of the connected data.

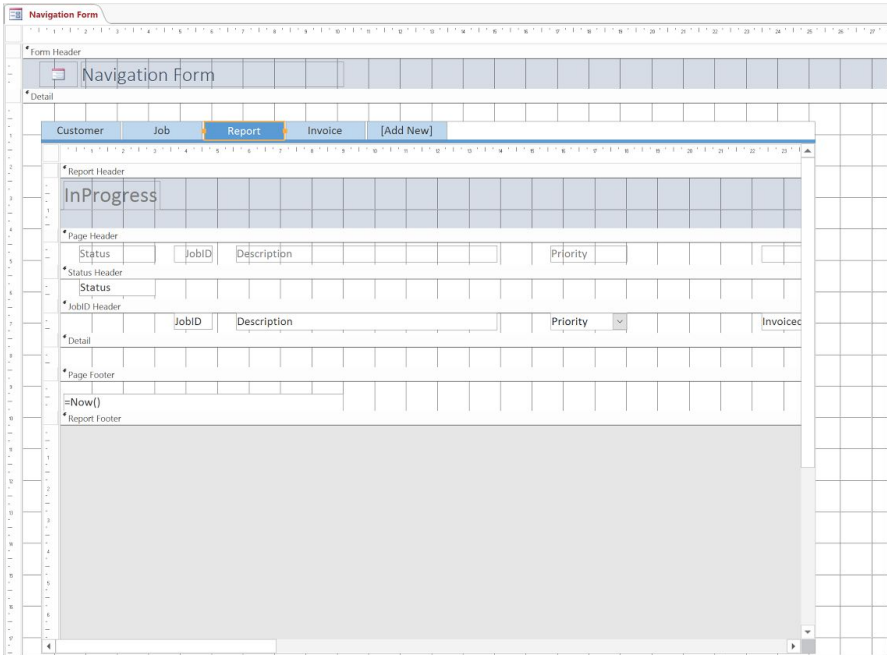


This is the design view of the Work form.

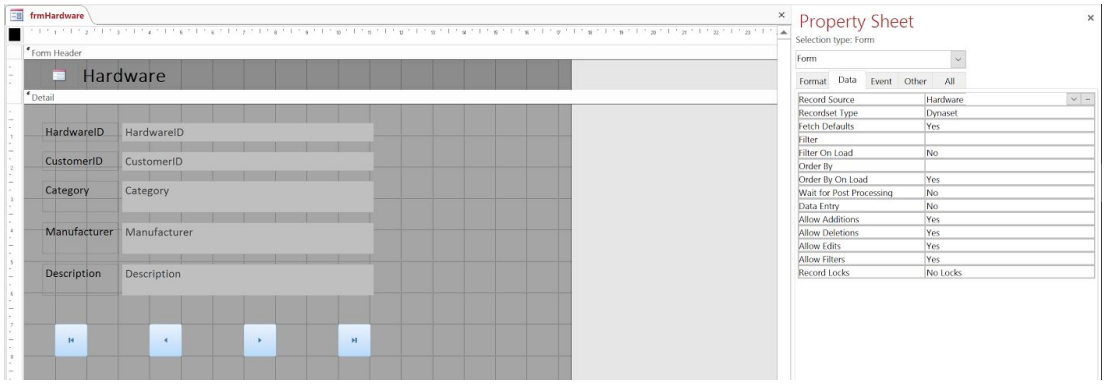

Navigation Form
Form Header
Navigation Form
Detail
Customer
Job
Report
Invoice
[Add New]
Form Header
Customer
Detail
CustomerID
CustomerID
CustomerFirstName
CustomerFirstName
Address1
Address1
Address2
Address2
Town
Town
frmHardware
Form Header
Hardware
Detail
HardwareID
HardwareID
CustomerID
CustomerID
Category
Category
Manufacturer
Manufacturer
Description
Description
Property Sheet
Selection type: Form
Form
Format
Data
Event
Other
All
Record Source
Recordset Type
Dynaset
Fetch Defaults
Yes
Filter
Filter On Load
No
Order By
Order By On Load
Yes
Wait for Post Processing
No
Data Entry
No
Allow Additions
Yes
Allow Deletions
Yes
Allow Edits
Yes
Allow Filters
Yes
Record Locks
No Locks


Navigation Form
Form Header
Navigation Form
Detail
Customer
Job
Report
Invoice
[Add New]
Report Header
HRR
54 small street
Page Header
CustomerFirstNar
CustomerLastNar
Address1
Address2
JobID Header
JobID
Manufacturer
Model
SerialNumb
Description
Detail
Action taken
Hourlyrate
HoursWorked
Cost
JobID Footer
=Sum([Cost])
Page Footer

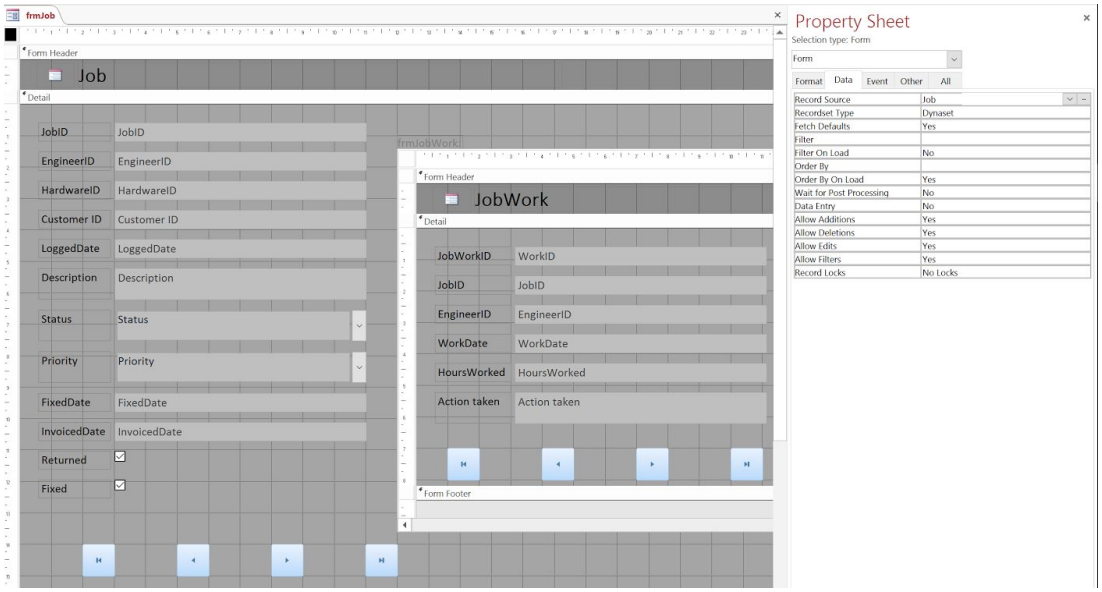




These screenshots show the design view of the Navigation form, which has all the forms which I have created. It contains the Customer form, which has the Hardware form embedded within the Customer form, it also has the Job form, which has the Work form embedded within the Job form, it has a report which shows the jobs which are in progress and an invoice section. The invoice section would request a Job ID number from the user, in which it will calculate the cost of the job.

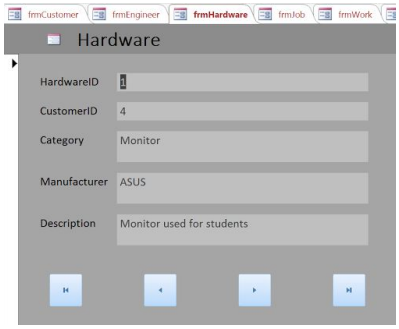


This is the design view of the hardware form.

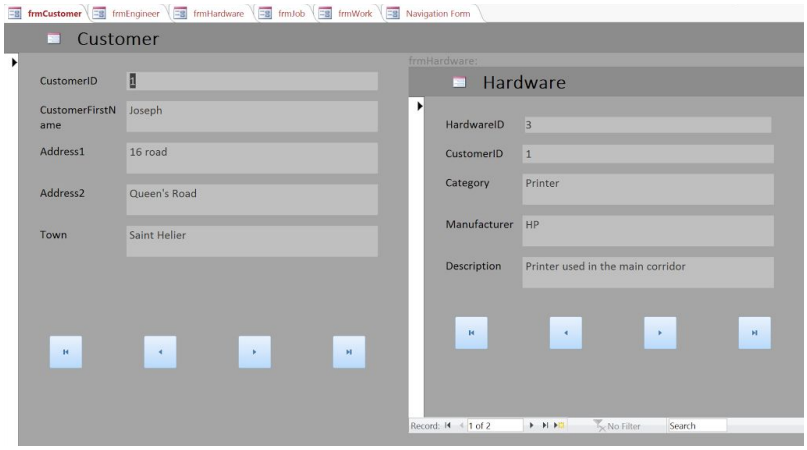


This is the design view of the Job form, which has the Work form embedded within the Job form, giving easier view of the connected data.

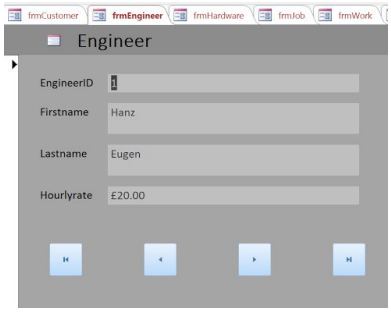
Your completed forms showing data



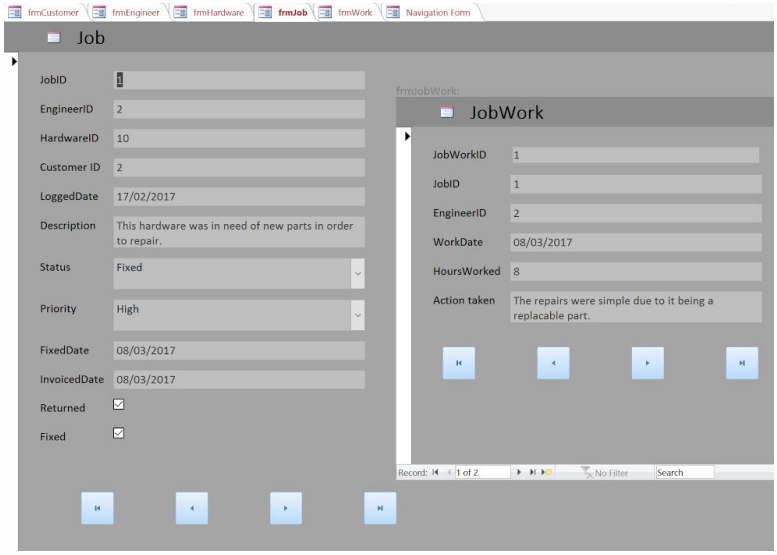
This is the hardware form showing the data added.



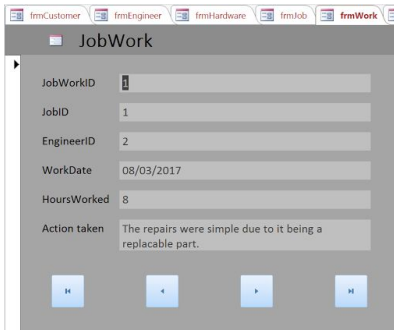
This is the customer form also showing the hardware form with both containing data.



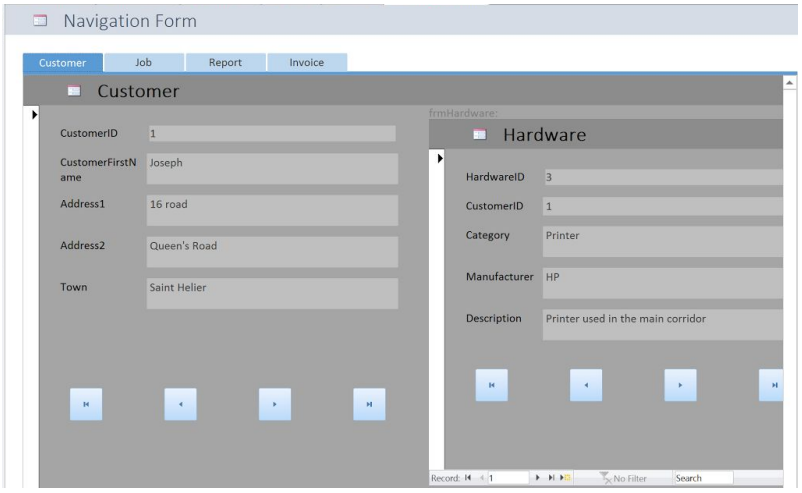
This is the engineer form showing the data added.

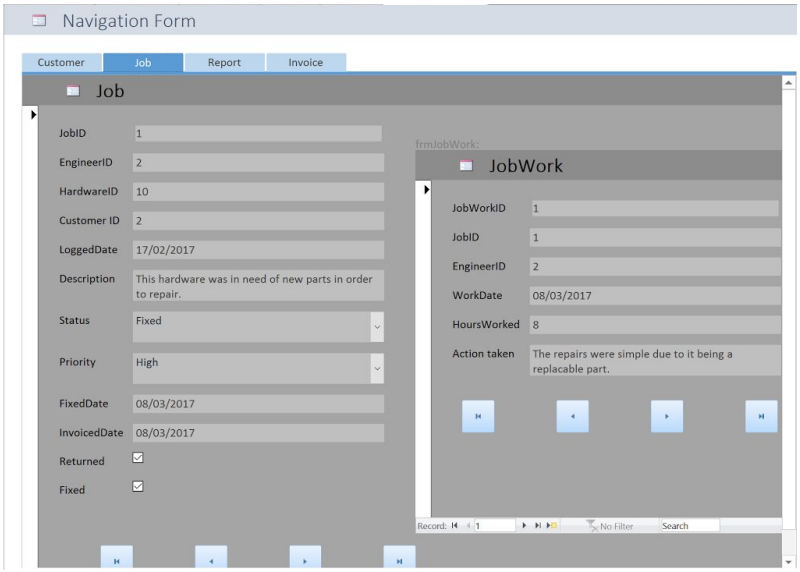


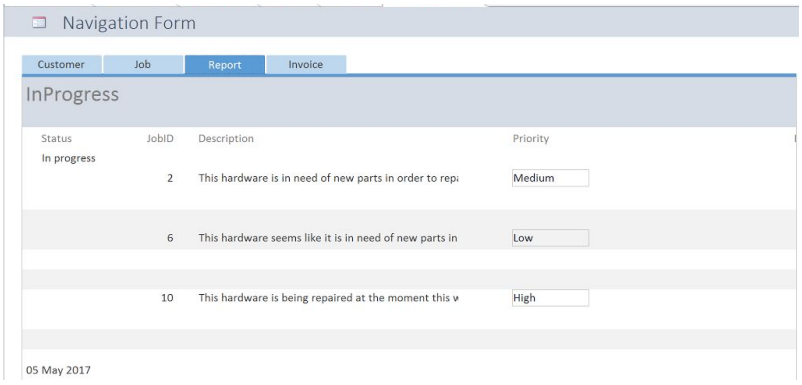
This is the Job form also showing the Work form (which I called JobWork) with both containing data.

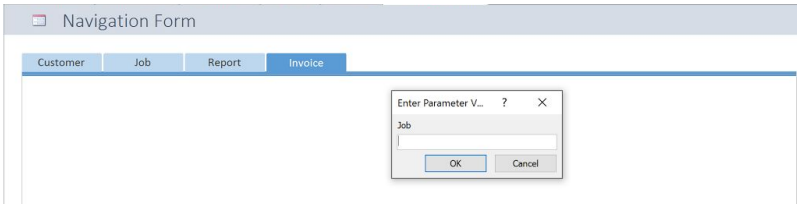


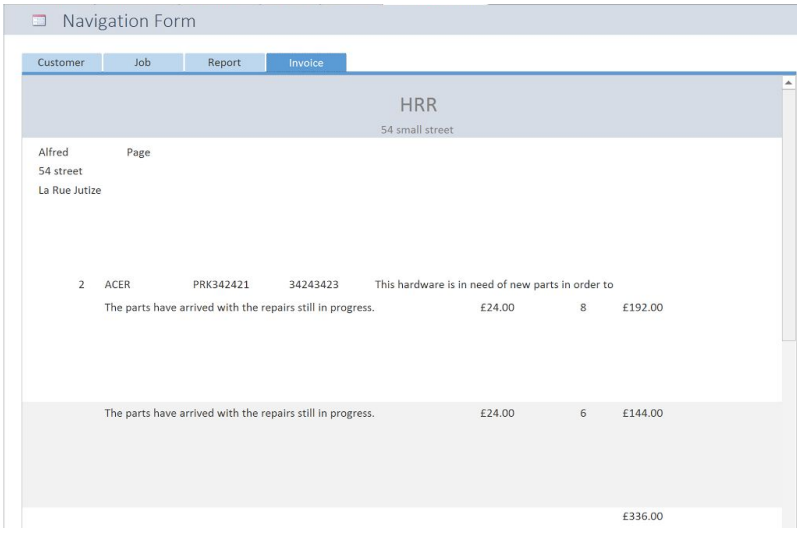
This is the Work form (which I called JobWork) showing the data added.







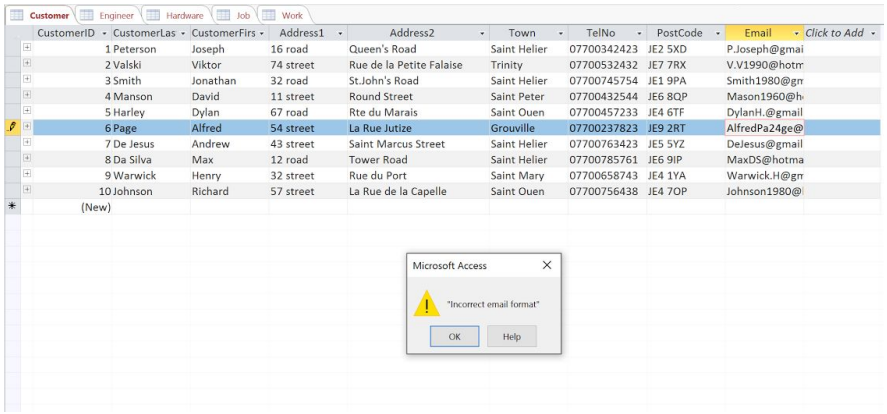




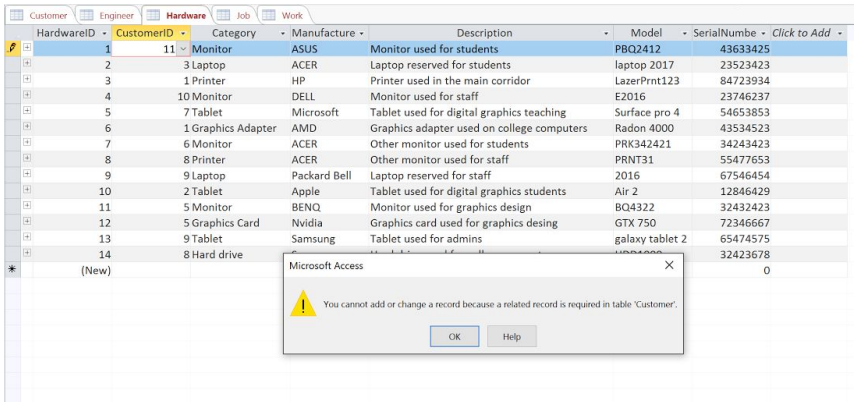
These screenshots show the Navigation form which, as I stated before, has all the forms which I have created. It contains the Customer form, which has the Hardware form embedded within the Customer form, it also has the Job form, which has the Work form embedded within the Job form, it has a report which shows the jobs which are in progress and an invoice section. The invoice section would request a Job ID number from the user, in which it will calculate the cost of the job.

<u>Your use of three DIFFERENT types of validation. Include screen-prints showing:</u>

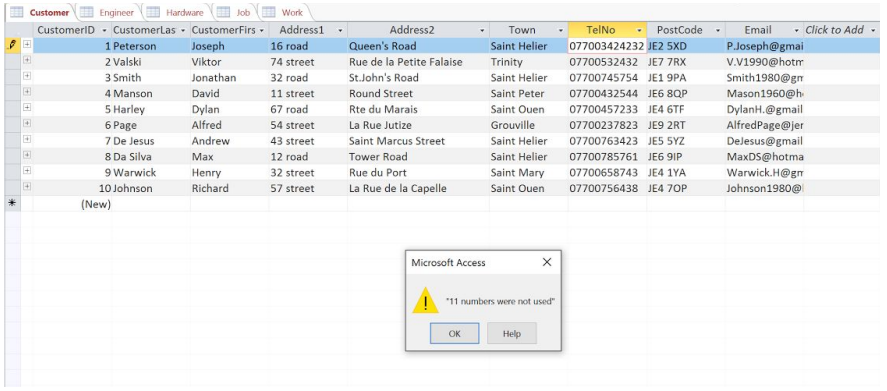
Invalid information and the validation error messages.



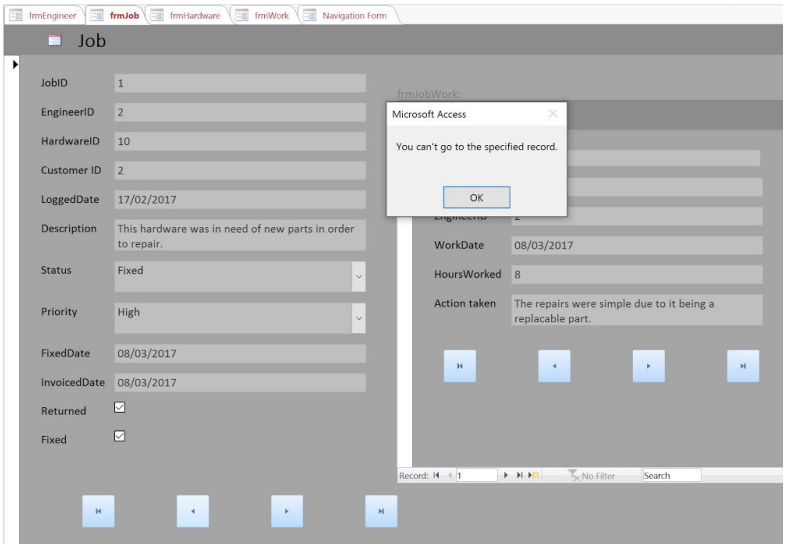
This is the error message that appears if the user put data which doesn't fit within the validation rule for the email.



This is the error message that appears if the user put data which doesn't exist on other related tables.

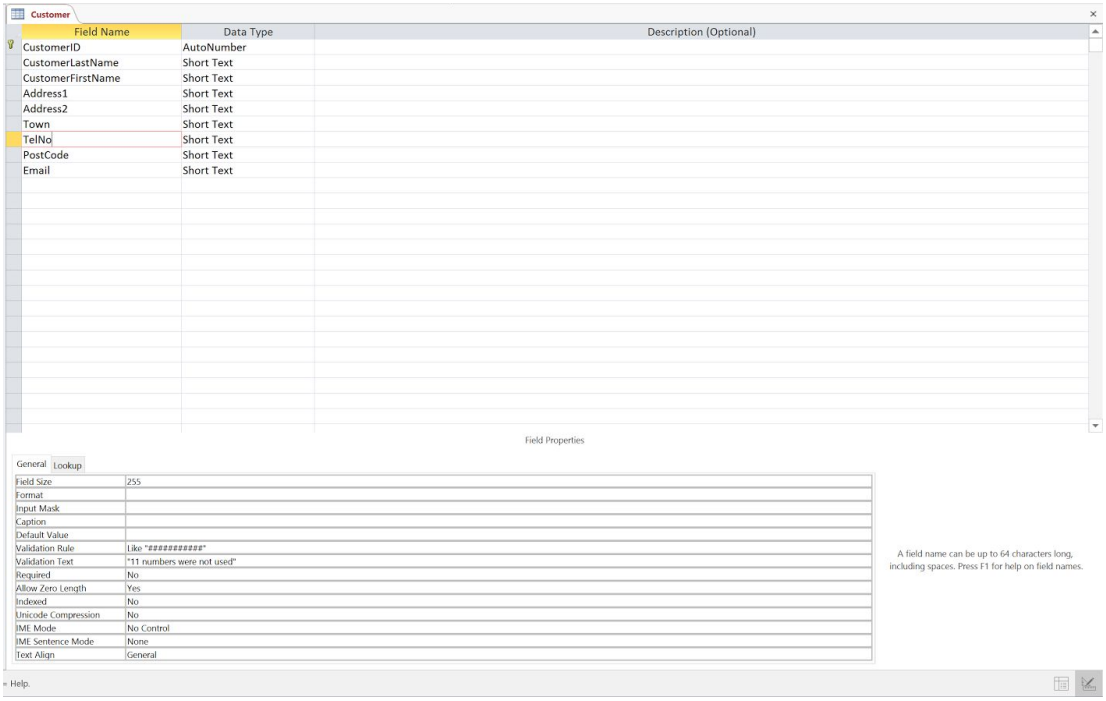


This is the error message that appears if the user put data which doesn't fit within the validation rule for the telephone number.

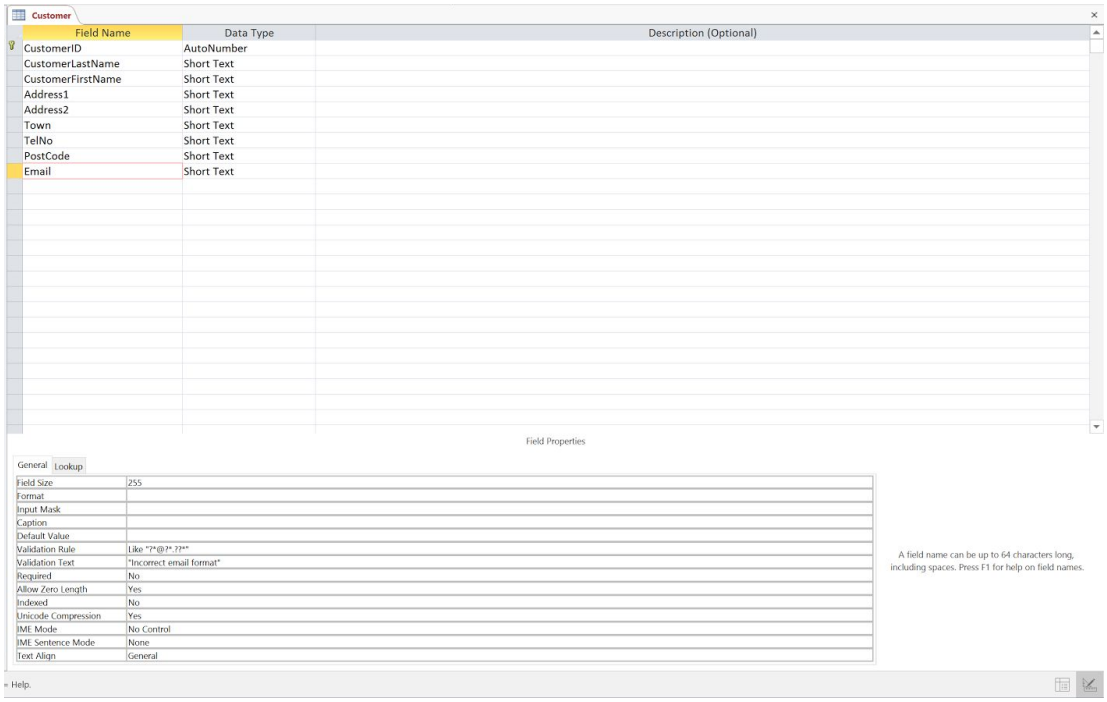


This is the error message that appears if the user tried and get information from a record which does not exist.

The relevant properties of the form (or table) in design mode.

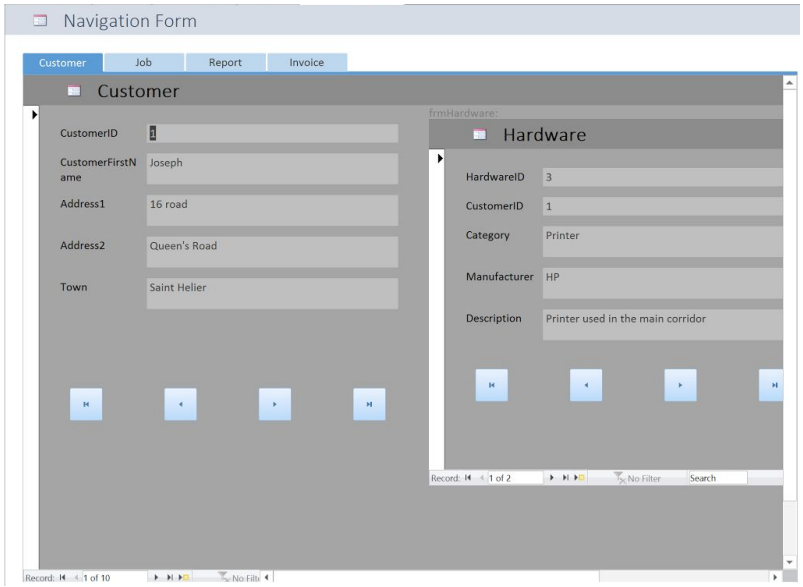


This is the design view for the validation rule and validation text for the telephone number attribute.

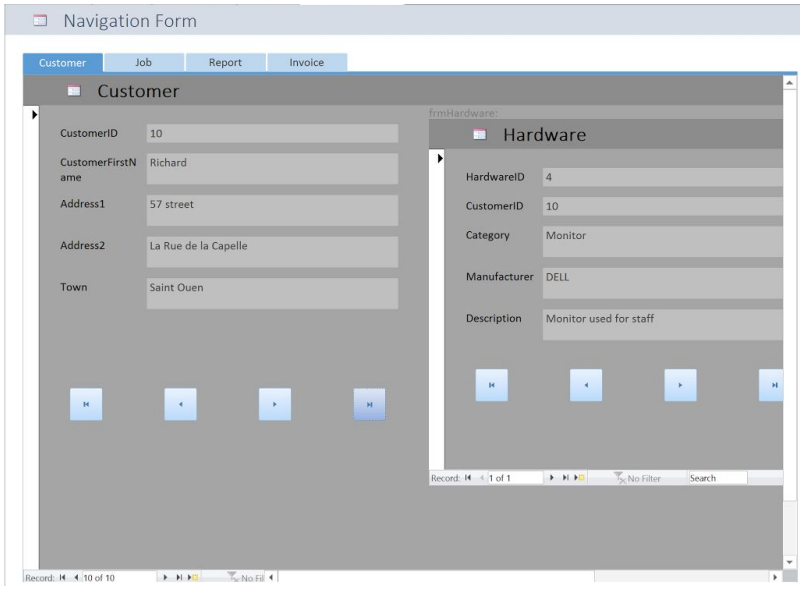


This is the design view for the validation rule and validation text for the email attribute.

Your use of an advanced feature (probably buttons) [U18P5].

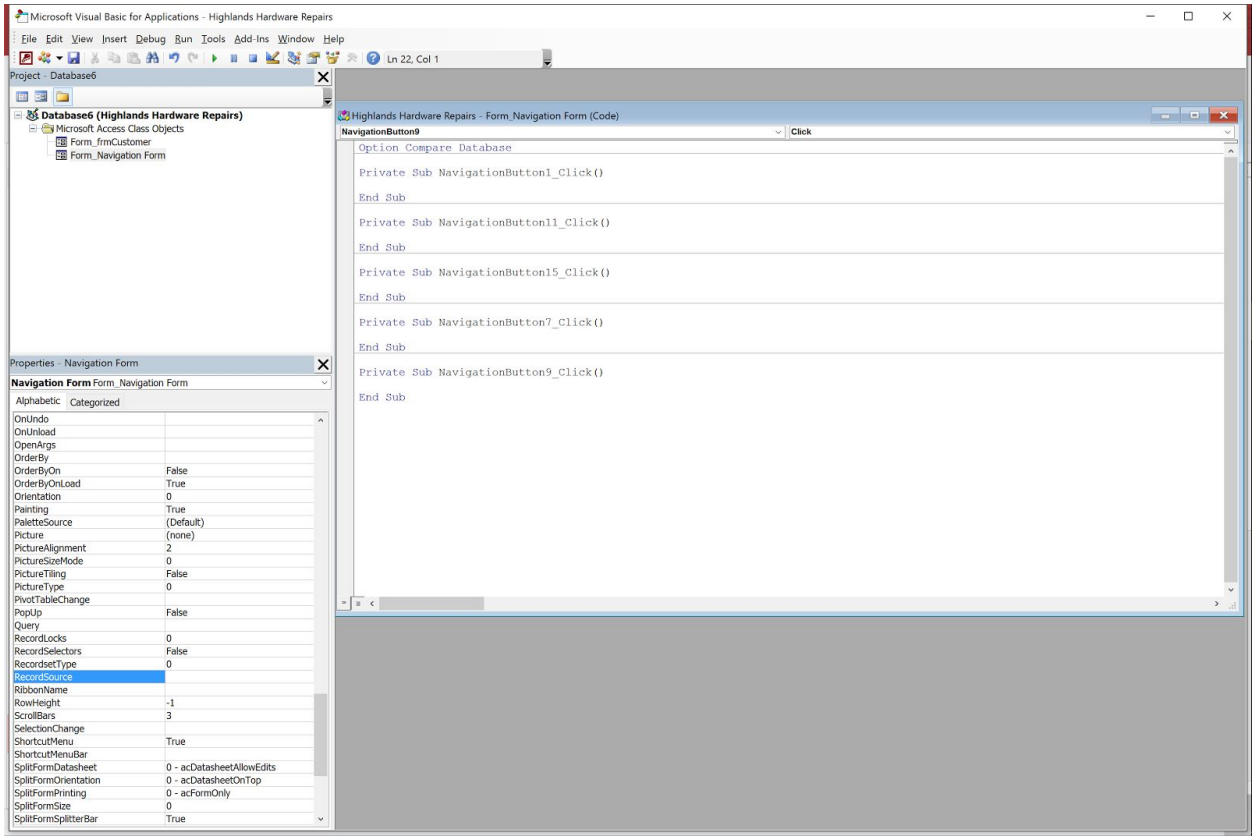


This is the navigation form before I pressed the button which goes the the last customer data.



This is the navigation form after I pressed the button to go the the last customer data.

Your program or macro code [U18M4].



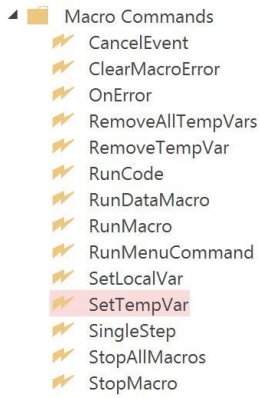
This is the code for the buttons on the navigation form.

Validation Rule	Like "##########"
Validation Text	"11 numbers were not used"

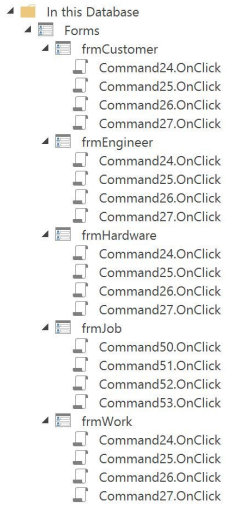
This is the validation rule and validation text for the telephone number attribute.

Validation Rule	Like "?*@?*.??*"
Validation Text	"Incorrect email format"

This is the validation rule and validation text for the email attribute.



This is the macro commands which is implemented within the navigation form.



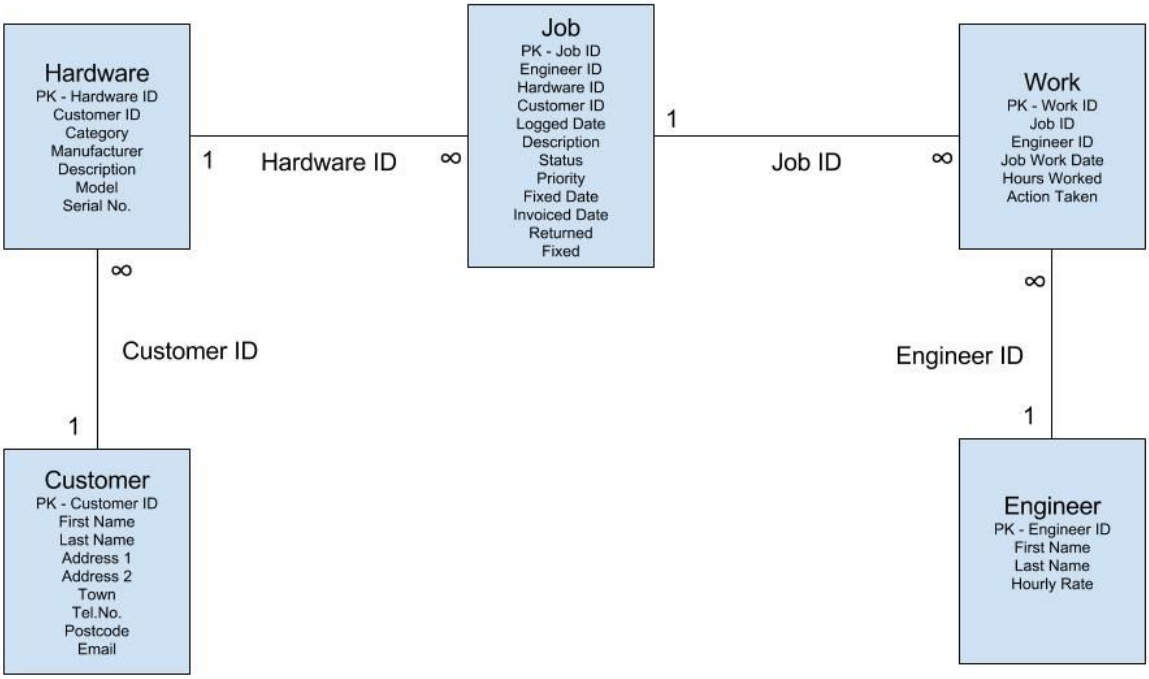
These are all the macro commands for the forms within my database.

Task 3

(U18:P2) - Design a relational database for a specified user need

Task description: *A design specification that includes:*

An entity relationship diagram



A data dictionary listing attributes need in each entity

Table Name	Attribute Name	Data Type	Purpose/Content	Validation
Customer	CustomerID	Autonumber	The purpose of this attribute is to make a unique ID number for a customer.	The validation to this attribute is the default validation for the data type.
	CustomerLastName	Short Text	The purpose of this attribute is to store the last name of a customer.	The validation to this attribute is the default validation for the data type.
	CustomerFirstName	Short Text	The purpose of this attribute is to store the first name of a customer.	The validation to this attribute is the default validation for the data type.
	Address1	Short Text	The purpose of	The validation to

			this is to store the address number of a customer.	this attribute is the default validation for the data type.
	Address2	Short Text	The purpose of this is to store the street address of a customer.	The validation to this attribute is the default validation for the data type.
	Town	Short Text	The purpose of this is to store the town name of a customer.	The validation to this attribute is the default validation for the data type.
	TelNo	Short Text (Validation Rule)	The purpose of this is to store the telephone number of a customer.	The validation to this attribute was added with a validation rule, which has “Like "###########"” written so that only 11 numbers are accepted.
	Postcode	Short Text	The purpose of this is to store the postcode of a customer.	The validation to this attribute is the default validation for the data type.
	Email	Short Text (Validation Rule)	The purpose of this is to store the email of a customer.	The validation to this attribute was added with a validation rule, which has “Like "?*@?*.??*"” written so that only a few characters with an “@” and a “.” are accepted.
Engineer	EngineerID	Autonumber	The purpose of this is to make a unique ID	The validation to this attribute is the default

			number for an engineer.	validation for the data type.
	Firstname	Short Text	The purpose of this attribute is to store the first name of an engineer.	The validation to this attribute is the default validation for the data type.
	Lastname	Short Text	The purpose of this attribute is to store the last name of an engineer.	The validation to this attribute is the default validation for the data type.
	Hourlyrate	Currency	The purpose of this is to store the rate an engineer is paid within 1 hour.	The validation to this attribute is the default validation for the data type.
	TelNo	Short Text (Validation Rule)	The purpose of this is to store the telephone number of an engineer.	The validation to this attribute was added with a validation rule, which has “Like "##########"” written so that only 11 numbers are accepted.
Hardware	HardwareID	Autonumber	The purpose of this is to make a unique ID number for hardware.	The validation to this attribute is the default validation for the data type.
	CustomerID	Number (Lookup Wizard)	The purpose of this is to store the automated ID of a customer to a specific hardware.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	Category	Short Text	The purpose of this is to store	The validation to this attribute is

			the type of hardware.	the default validation for the data type.
	Manufacturer	Short Text	The purpose of this is to store the manufacturer of a piece of hardware.	The validation to this attribute is the default validation for the data type.
	Description	Short Text	The purpose of this is to store what the hardware is used for.	The validation to this attribute is the default validation for the data type.
	Model	Short Text	The purpose of this is to store the model the piece of hardware is.	The validation to this attribute is the default validation for the data type.
	SerialNumber	Number	The purpose of this is to store the serial number of the hardware.	The validation to this attribute is the default validation for the data type.
Job	JobID	Autonumber	The purpose of this is to make a unique ID number for a job.	The validation to this attribute is the default validation for the data type.
	LoggedDate	Date/Time	The purpose of this is to store the date that the hardware was logged.	The validation to this attribute is the default validation for the data type.
	Description	Short Text	The purpose of this is to store what the hardware needs in order for a repair.	The validation to this attribute is the default validation for the data type.
	Status	Short Text	The purpose of	The validation to

		(Lookup Wizard)	this is to store whether the hardware is “fixed”, “in progress”, “unrepairable”, “awaiting parts” or “binned”.	this attribute is that it only accepts data that has been added through the lookup wizard.
	Priority	Short text (Lookup Wizard)	The purpose of this is to store the priority of the hardware, whether it is “high”, “medium” or “low”.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	EngineerID	Number (Lookup Wizard)	The purpose of this is to store the automated ID of an engineer to a specific job.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	HardwareID	Number (Lookup Wizard)	The purpose of this is to store the automated ID of a piece of hardware to a specific job.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	Customer ID	Number (Lookup Wizard)	The purpose of this is to store the automated ID of a customer to a specific job.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	FixedDate	Date/Time	The purpose of this is to store	The validation to this attribute is

			the date that the hardware was repaired.	the default validation for the data type.
	InvoicedDate	Date/Time	The purpose of this is to store the date that the hardware was invoiced.	The validation to this attribute is the default validation for the data type.
	Returned	Yes/No	The purpose of this is to store whether the hardware was returned or not.	The validation to this attribute is the default validation for the data type.
	Fixed	Yes/No	The purpose of this is to store whether the hardware was fixed or not.	The validation to this attribute is the default validation for the data type.
Work	WorkID	Autonumber	The purpose of this is to make a unique ID number for work.	The validation to this attribute is the default validation for the data type.
	JobID	Number (Lookup Wizard)	The purpose of this is to store the automated ID of a job to a specific work.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	EngineerID	Number (Lookup Wizard)	The purpose of this is to store the automated ID of an engineer to a specific work.	The validation to this attribute is that it only accepts data that has been added through the lookup wizard.
	WorkDate	Date/Time	The purpose of this is to store	The validation to this attribute is

			the date that the work started.	the default validation for the data type.
	HoursWorked	Number	The purpose of this is to store the time it took or is currently taking for the hardware to be repaired.	The validation to this attribute is the default validation for the data type.
	Action taken	Short Text	The purpose of this is to store the action that was taken in order to repair the hardware.	The validation to this attribute is the default validation for the data type.

Task 4

(U18:P7) - Test a relational database

Task description - *Create a test plan in the following form:*

Test Number	Action	Expected result	Actual Result & Comments *(just tick if successful)*

The scanned copy of the test plan you completed for a colleague

Test Number	Action	Expected result	Actual Result & Comments *(just tick if successful)*
1.	Test referential integrity *Try to add a record to the many side of an enforced relationship without there being a matching record on the one side*	In this test it is expected that the record added on the enforced relationship without there being a matching record on the one side.	✔
2.	Test referential integrity *Try deleting the record*	In this test it is expected that the record with the	✔

	on the one side of an enforced relationship where there are records on the many side and cascade delete has not been enabled	cascade delete doesn't enable the deleted record on the side of an enforced relationship.	
3.	Test normal data entry *Ensure that properly formatted information can be entered into the customer, hardware, job and work record tables using your forms*	I expect that the data can be entered properly in all the tables.	✔
4.	Test normal data retrieval *That the records entered in the last bullet point can be retrieved once the forms have been cleared*	I expect that the data can be retrieved once the form have been cleared.	✔
5.	Test extreme and erroneous data validation *Test to make sure that invalid data cannot be entered.*	I expect that the data can't when invalid.	✔
6	Test more extreme and erroneous data validation *Test to make sure that valid data can be entered*	I expect that the data can be entered when valid	✔
7.	Test buttons *Make sure that all the buttons on the your forms work as expected*	I expect that the buttons on the database to work properly.	✔
What do you like about the database? What I liked about the database is that it has worked properly without there being an unexpected error or the database crashing. I think that the functionality of the database is very good and that it has good amounts of data.			
What do you think should be improved? What I think should be improved is the visuals of the records that you made in the database because it all looks like it doesn't have a lot of quality and that the colours you added is the only visual improvement you made.			

Name Dylan F	Signature DF	Date of testing 03.05.2017

The scanned copy of the test plan your colleague completed for you

Test Number	Action	Expected result	Actual Result & Comments *(just tick if successful)*
1.	Buttons	I expect that the buttons on the database to work properly.	✔
2.	Data entry	I expect that the data can be entered properly in all the tables.	✔
3.	Data retrieval	I expect that the data can be retrieved once the form have been cleared.	✔
4.	Data validation	I expect that the data can be entered when valid and can't when invalid.	✔
5.	Integrity	I expect the data within my database to be consistent and that the data that is referenced to be valid.	✔

Things you like about the database?

The things I liked about the database is how well it shows data and also how well it went throughout this test. I believe that the database you made is very refined and that it should be a completed database in which a customer can use to store, validate, navigate and use the data throughout.

Things you think should be improved?

The things I think that should be improved in your database is the aesthetics of the reports. I feel that the design of the reports are too crude and need better visual improvement. However, I feel that it is the only improvement since the database works well without any issues.

Name	Signature	Date of testing
Leandro Baroca	L.B	03/05/2017

Task 5

(U18:D2) - Evaluate a database against the specified user need

Task Description - *Evaluate your database by creating a one to two-page report that discusses:*

Discussion of how well the business needs are met

I feel that the database that I have created meets the intended purpose well. The reason for this is because the database gives a good amount of information on what the user would want to access to find out the data of the business.

In the design specification, it was specifically said that the application has to be populated with at least 5 correctly related tables, have some sort of data validity, integrity of data and allows the user to navigate easily with an automated function.

With the 5 correctly related tables, I implemented a lot of data in the tables which tells the user which customers, engineers, hardware, jobs and work that is required within the business in order to carry out the business's intended purpose.

With the data validity and integrity of data, I implemented a few macros in the design view which gives a restriction on what type of data can be written. If the data written was wrong, the database will give an error message which indicated what the user did wrong.

The validation methods that I wrote myself gives restriction on data to the user on what data should be written.I felt that this would make it easier to the user so that the data would have a less chance of being incorrect.

With the automated function, I believe that this is very important to have because , what I feel, that is what makes the customer use this database for. For instance, after seeing all the data, the user will see what data is connected with each other, which I feel is the only reason that this database was made.

Also, the automated function gives ease of navigation and how to access the data that is needed, however, I feel that there could be more automated functions which could help improve the user friendliness of the database.

Overall, I feel that the database that I created does carry out its intended purpose that I have set in the design specification. However, I feel that this database can be vastly improved due to its very simple and crude nature. Improving aspects such as a better code for automatic functions, making it more aesthetically pleasing and other improvements that make it much more user friendly which would make the database a lot easier to use.

Discussion of differences between the design and actual completed database

There are very little differences between my design and the actual completed database. I would say that the design is very identical to the completed database due to the fact that I have planned everything before creating the actual database itself. However, even though it is very identical, it isn't completely identical.

There are a few differences in the completed database when compared to the design. For instance, the relationship diagram was different on the design task when compared to the completed one. However, the relationship diagram which I initially designed was incorrect so it had to be changed.

The reason that the relationship diagram was different was due to the data in the initial design being the incorrect type of data that is meant to be used within the database. So, I had to change the data in the design relationship diagram so that the data is correct to the purpose of the database.

Another difference in the completed database when compared to the design is the data that is included within the database. Due to the database requiring modification as I was making it, I had to delete previous data and make data which fit into the validation rules in order to have more accurate data.

The reason that the data was changed was mostly due to the validation rules, especially the telephone number requiring 11 numbers. In order to create the validation rules I had to delete the previous data, create the validation rule and recreate the data within the table.

In conclusion, these were the only differences that the completed database had compared to the design. These differences are really insignificant, however, these were the very few differences the database had to the design.

Discussion of good features in your database and why they are beneficial

I feel that my database has good features in which they benefit the user when it is being used. I feel that the features that I have made which are very beneficial is the use of related tables, the controls on my forms, the use of my validation and the reports in which I have made.

For the use of related tables, I feel that having these tables being related makes it much easier to see the data that can be related with each other. This is beneficial because it makes data relate to each other, giving me the ability to include useful information which can be located in one table, to be in a different table where it can be equally as useful.

In my database, the related tables have the ID of the table, such as a customer, as a primary key. Which is used in different tables where it would be useful to see which data goes with which. For example, the user will get to see which customer has what piece of hardware or what hardware is going to be repaired by which engineer.

For the use of controls on your forms, I feel that the forms help with the gathering of information a lot easier and faster, which is very beneficial to a user looking for information with very little time. Also, it allows people to calculate the costs with a lot more ease, giving users the ability to view, calculate a cost and pay to the business.

In my database, the forms contain data which all the tables contain and an extra form which is the navigation form. These forms all have buttons to navigate through the data, which the user can find out what data means what. With the navigation form, the user can use it to find out which customer has what hardware, which job has which work, a report showing which parts are in progress and an invoice which he can choose from the desired job to find the total cost.

For the use of validation, I feel that having this is a benefit because it gives much more accurate data. The reason for this is due to the validation restricting data that isn't valid for the type of data that is supposed to be stored.

In my database, having validation rules in place allowed the business to be sure that the data that contains the contact details of the customers and engineers to be correct, which is very important if a business wants to contact either a customer or an engineer for an important message.

For the reports, I feel that they are beneficial to a business because it shows important data to a user who wants to find out that specific data. What I mean by this is that the user can use the reports in the database to find out which data has a higher priority than others or whether the user wants to use the report to also calculate the cost.

In my database, the reports consist of data which contain which I feel is the highest priority of data. Which is hardware which is fixed/unrepairable, hardware that is in progress, hardware that is in progress with a high priority and the invoices which is used to calculate the cost of a repaired hardware.

In conclusion, I believe that the features that I have stated above are very beneficial to the business because it makes accessing data much more easier and faster for the user, so the user won't have to look for data and get frustrated or confused. These features also gives other benefits such as the ability to separate the high priority jobs from the lower priority ones and the ability to calculate the cost of the repaired hardware.

<u>Discussion of possible improvements and why they would help the business</u>

I feel that it is possible to have improvements that could be implemented within my database. The reason that I think this is because my database is nowhere near perfect due to my lack of database knowledge, so it is always possible to have an improvement to my database.

The improvements which I believe can help the business is an improvement to the data model

(changes to or additional tables and fields), changes to existing forms, additional forms, changes to existing reports and additional reports.

For the improvements to the data model, I believe that I could improve the data model by adding more data so that the database could be used to its limit, so that more improvements can be found and that the forms and reports will be more populated in data.

Also, another improvement which I feel should be implemented within the data model, is having more attributes. This is so more useful data can be stored which can be used to prioritise data more and to include other important data to use.

This would help the business because with a lot more data, it would increase the chance of an error occurring on the database. This may seem like a disadvantage, but it means that a previous unkown error is found and that it could be repaired, making it a much more refined database.

For the changes to existing forms, I believe that I could improve them by having a better navigation and to make it much more aesthetically pleasing to the user. The way I can improve this is by making a switchboard, which is basically a way of navigation which the user would click a specific button to access data.

This would help the business because a switchboard will give the ability for the user to click on a specific button, which will take the user to a specific form. I feel that this is beneficial because it makes it easier for the user to find the specific data that the user needs, without the user being populated by information which would be deemed useless to the user.

For the additional forms, I believe that I could add more forms which focus on higher or lower priorities. For instance, I could implement forms which has hardware which has a status on waiting for parts with the engineers contact details so that they can be called to say whether the additional parts have arrived or not.

This would help the business because it could improve the knowledge the business has on the data and why some of the hardware repairs have been postponed. I feel that this is beneficial because it makes it much easier for the company to track how long it will take for the hardware to be repaired, so that they can give somewhat accurate information to the customer waiting for the hardware.

For the changes to existing reports, I believe that I could improve them by having a better view of the data and to make it much more aesthetically pleasing to the user. The way I could improve this is by changing all the text and the size of the boxes in the design view of the reports, so that it would be easier to pinpoint useful data without having to read through unneeded data.

This would help the business because the user would have to look through the data that is applicable for the situation. What I mean by this is that, since a report is printed and given to a manager, not only will it have to contain useful data, it would also have to have an easier way to direct to the data that is needed. So having visual improvements and guidance makes it easier to read the data needed.

For the additional reports, I believe that there could be additional reports which focus on other priorities and the date the hardware is being repaired. I feel that this is beneficial because it can contain data which shows the date it started and other priorities so that they can still be used when compared to other data which has a lower or higher priority.

This would help the business because it would give the business the ability to compare the different dates and priorities each customer and hardware has. The way that this is an improvement is due to the fact that priorities can be changed by the manager, telling me the priority of a job due to the time it took to repair being too long for the customer.

Conclusion

In conclusion, I believe that my database has met its initial purpose well, which is to store and show data to the user about the hardware repairs. However, I feel that there is plenty of room for improvement in my database.

With the database being a complex application to use, it gives it the ability to be customised in various ways, but it will require a lot of experience. This means that I can tailor the database to the user's needs and further improve the database so it is more aesthetically pleasing and easier to use due to better navigation.

Overall, I feel that my database is at a very good stage due to its ability to contain and show data to a user. However, it is nowhere near perfect. Which is why I want to bring the database as close to perfect as I possibly can.